

Panum Dome Volcanic Map and Eruptive History

Motivation

Mapping the surficial geology of Panum Crater provides an opportunity to determine 1) the sequence of dome eruption and 2) their associated textures. Additionally, 3) the efficacy of volatile degassing during magma ascent can lend insight into 4) the effusive-explosive eruptive nature of Panum Crater.

Mono-Inyo Volcanic Chain Regional Context

Walker Lane and the associated Sierra Nevada range front fault system are a byproduct of dextral motion between the Pacific plate and Basin and Range province extension on the North American Plate. Steep normal faulting and strike slip shearing are the most common structures within the fault system and are hypothesized to accommodate microplate translation westward (Unruh, 2003). Long Valley Caldera sits on this boundary and has an ~ 3 Ma history of mafic to dacitic volcanism. E-W extension and subsidence of the Long Valley Caldera allowed for the intrusion of N-S oriented dikes and are theorized to be responsible for the formation of the Mono-Inyo volcanic chain ca. 50-35 ka (Black et al. 2016, Reches and Fink, 1988).

Recent volcanic activity along the northern end of the Mono-Inyo chain over the last ~600 years have produced rhyolitic domes and deposits, termed the North Mono Eruption. The eruption of Panum Crater (Fig. 1) is hypothesized to have resulted from the intrusion of a N-S oriented dike beneath the northern 6 km of the volcanic chain, in response to elastic rebound of the lithosphere from crustal extension (Wohletz, 1987, Sieh and Bursik, 1986). Magma was extruded in the subsurface below groundwater saturated alluvial deposits producing hydrovolcanic, sub-Plinian to Plinian eruptions ranging from effusive to explosive. The initial phreatic eruption produced the North Mono Airfall tephra deposits and was followed by a throat clearing breccia eruption or massive unsorted breccia, lapilli and accidentals sediments and Sierran clasts. Pyroclastic flow and surge deposits followed, emplacing beds of lapilli

gravels, laminated pumice and planar to cross-bedded ash containing lapilli and block bombs, common of the SW dome (Fig.1, Sieh and Bursik, 1986). The triangular facet observed on the west side of the tephra ring is likely associated with these initial pyroclastic eruptions and was later covered by tephra of the younger North Dome eruption. A subsequent massive phreatic explosion excavated meter scale glassy rhyolitic blocks of breccia and bread-crust pumice (Fig. 2E) from the inner crater, collapsing it and producing the Panum Block-and-Ash-flow deposit, preserved to the N-NW of Panum Crater. A pyroclastic flow deposits a layer of ash and lapilli, before a final massive eruption extruded aphanitic flow-banded obsidian (Fig. 2F), pumice and stony rhyolite (Fig. 2B), producing the North Dome of Panum and its associated tephra ring (Fig. 1, Fig. 2P). Relatively small dike widths (<3 m) have led Sieh and Bursik (1986) to the hypothesis that magma mixing in conjunction with tectonic strain were likely responsible for the over-pressurization of the Mono Crater magma reservoir leading to the eruption and emplacement of Panum Dome. The lack of lacustrine and aeolian sedimentary beds within the pyroclastic beds imply the eruption occurred over a few months (Sieh and Bursik, 1986).

Eruptive Sequence and Unit Descriptions

Panum Crater is composed of four main domes: a Southwest, East, Southern and North Dome. The Southwest and East domes seem to be stratigraphically the oldest and are hypothesized to be fragments of previous domes, pushed to the flanks of the crater by the eruption of the Northern and South Domes. The smaller South Dome forms a graben that displays N-S bilateral symmetry likely from E-W extension in response to the upward intrusion of the younger North Dome dike into the fissure of the South Dome. Boudinage of flow banding (Fig. 2I) often captures this extension. The South Dome is older than the North Dome, since it is bisected by the southernmost tip of North Dome's eruptive fissure (Fig. 1).

The Southwest dome is composed of plentiful large blocks of cream-colored stony rhyolite (Fig. 2A) that outcrop along the western flank of the South dome. The East dome also displays a similar color and texture of stony rhyolite that grades to pumiceous rhyolite (Fig. 2B) up stratigraphy and is surrounded by a thin band of tephra locally, from which Pele's Tears

(Fig. 2G) can be sourced. The increased pumicity preserved in the stony rhyolite of the East dome may imply a slightly younger age than the Southwest dome. Both domes are highly fractured and brecciated, hypothesized to occur during emplacement of the North Dome (Sieh and Bursik, 1986). The South Dome displays frequent occurrences of massive microvesicular stony rhyolite and banded obsidian (Fig. 2A,D) beneath a 10-meter sheath of foliated pumice (Fig. 2B) coating in a meter thick obsidian shell (Fig. 2D, Sieh and Bursik, 1986). The largest and youngest dome of the crater is the North Dome which comprises 2/3 of Panum's exposed surface and is chiefly composed of large blocks of interbedded pumiceous rhyolite and thin, planar bedded obsidian (Fig. 2B).

Along the northern most side of Panum Crater the talus that composes the sides of the dome increases in size from cobble to boulder, suggesting a massive pyroclastic eruption to the north of the North dome sent. Near vertical beds of outwardly dipping flowbanded stony rhyolite (Fig. 2A) with a breadcrust rind (Fig. 2E) are common on crease structures (Fig. 2M) parallel to the N-S fissure that bisects the extent of the North Dome. Backfill breccia (Fig. 2C) can also be seen within the central Northern fissure and in structurally coherent spines. Significant auto-brecciation (Fig. 2J) of the fissure walls and spines from magma injection and volatile expansion can be observed. Tuffisite veins are also common at the southern end of the North Dome parallel to the fissure and within the South Dome on either side of the graben. Fumaroles and fumarolic alteration (Fig. 2K) are common along the fissures and the flanks of the four domes.

Devolatilization of Melt

Spatter cones, fumarolic alteration, bread crusts, massive obsidian blocks and the vesicularity of rhyolite all record the efficacy of volatile degassing events within the melt during the eruptions of Panum Crater. Spatter cones (Fig. 2N) and vents found within the crater (best observed just east of the South Dome's fissure and graben structure) support simultaneous effusive and explosive eruptive behavior and are often coupled with localized tephra cones that contain Pele's Tears; small bits of phreatic ejected melt that rapidly cool as rhyolitic pumice while volatiles escape forming a breadcrust rind (Fig. 2E,G). The massive oxidation marks produced

from fumarolic alteration record intense vent degassing events, where large quantities of gaseous water interact with iron present in the volcanics to stain them red-orange. Rapid evacuation of volatiles from a quickly depressurized and extruded melt can produce the massive blocks of obsidian that are commonly exposed within the graben and fissure of the South Dome, where some devitrification textures can be appreciated (Fig. 2H). The size of rhyolite pumice vesicles directly correlates to their amount of degassing and stratigraphic location. Within Panum domes frequently display massive obsidian at their bases, overlaid by interbedded microvesicular stony rhyolite and obsidian, which increases in vesicles upward to pumiceous rhyolite (though the percentages of obsidian are quite variable locally). At Panum, phreatic volatile degassing eruptions were frequently accompanied by small scale effusive eruptions of thick felsic-rich rhyolitic magmas.

Hybrid Effusive-Explosive Eruptive Evidence

Tuffisite (Fig. 2L) and backfill breccia (Fig. 2C) and pyroclastic bombs (Fig. 2O) trapped in flows record the simultaneous effusive-explosive eruptive style of Panum Crater. Ejecta extruded from the vents at angles greater than 45° frequently fall back into the fissure, and are thus reincorporated into newly extruding, effusive flows. The obsidian and ash fill cracks in the breccia and amalgamate fractured rhyolitic clasts together creating a backfill breccia structure. Similarly, large spires formed from effusive flows begin to crack and joint as soon they extrude, allowing tuffisite (Fig. 2L) deposition as ejected clasts and ash are blown into the cracks produced by this auto-brecciation. Pyroclastic bombs trapped in flows (Fig. 2O) also record synchronous magma extrusion and phreatic eruptions. Additionally, the presence of a tephra ring around the East Dome and the presence of micro to macrovesicular stony rhyolite associated with the South Dome beneath the western ring of the tephra cone provide strong evidence of consistent ash venting and effusive eruptive styles.

Figure 1 – Surficial Map

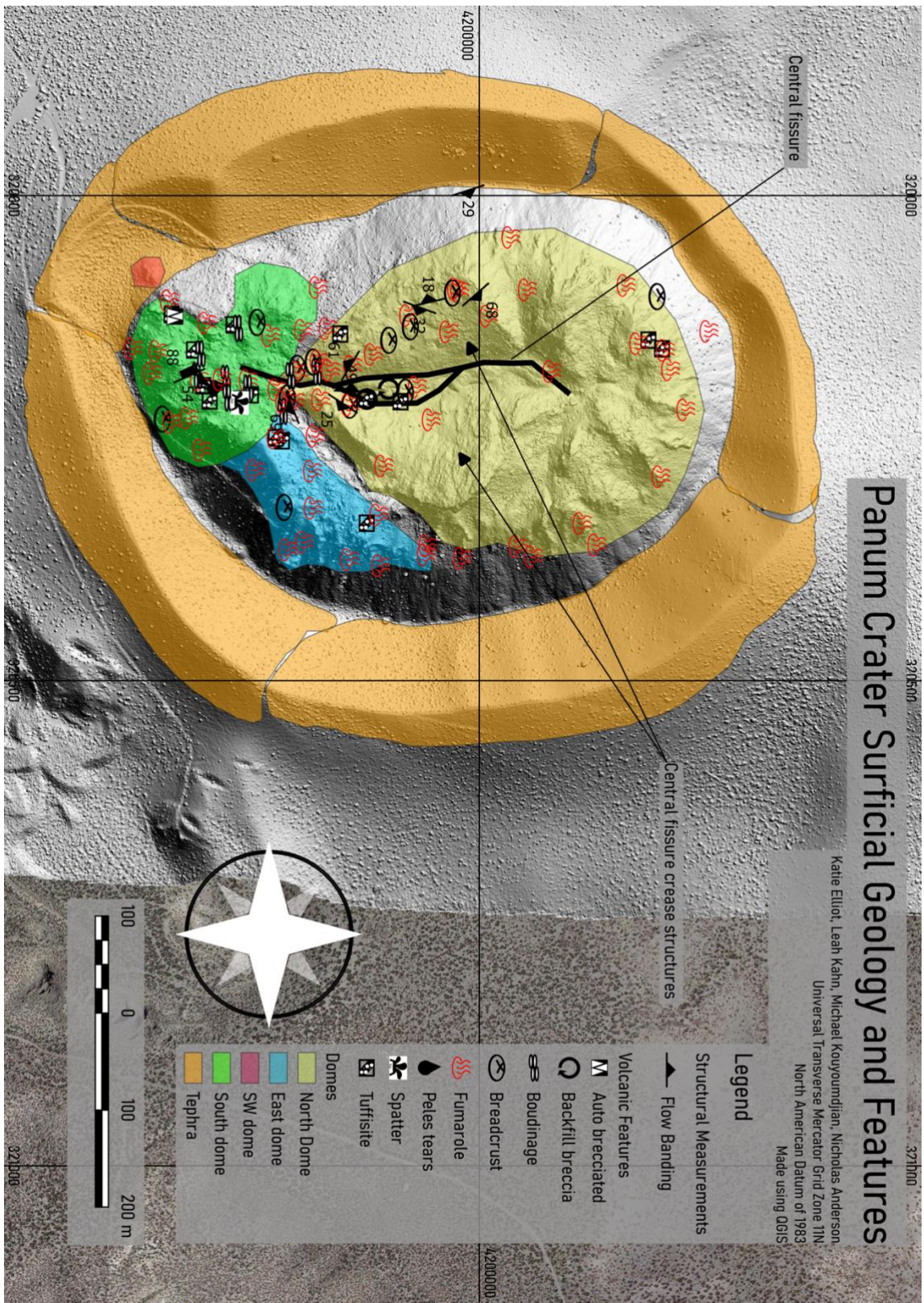
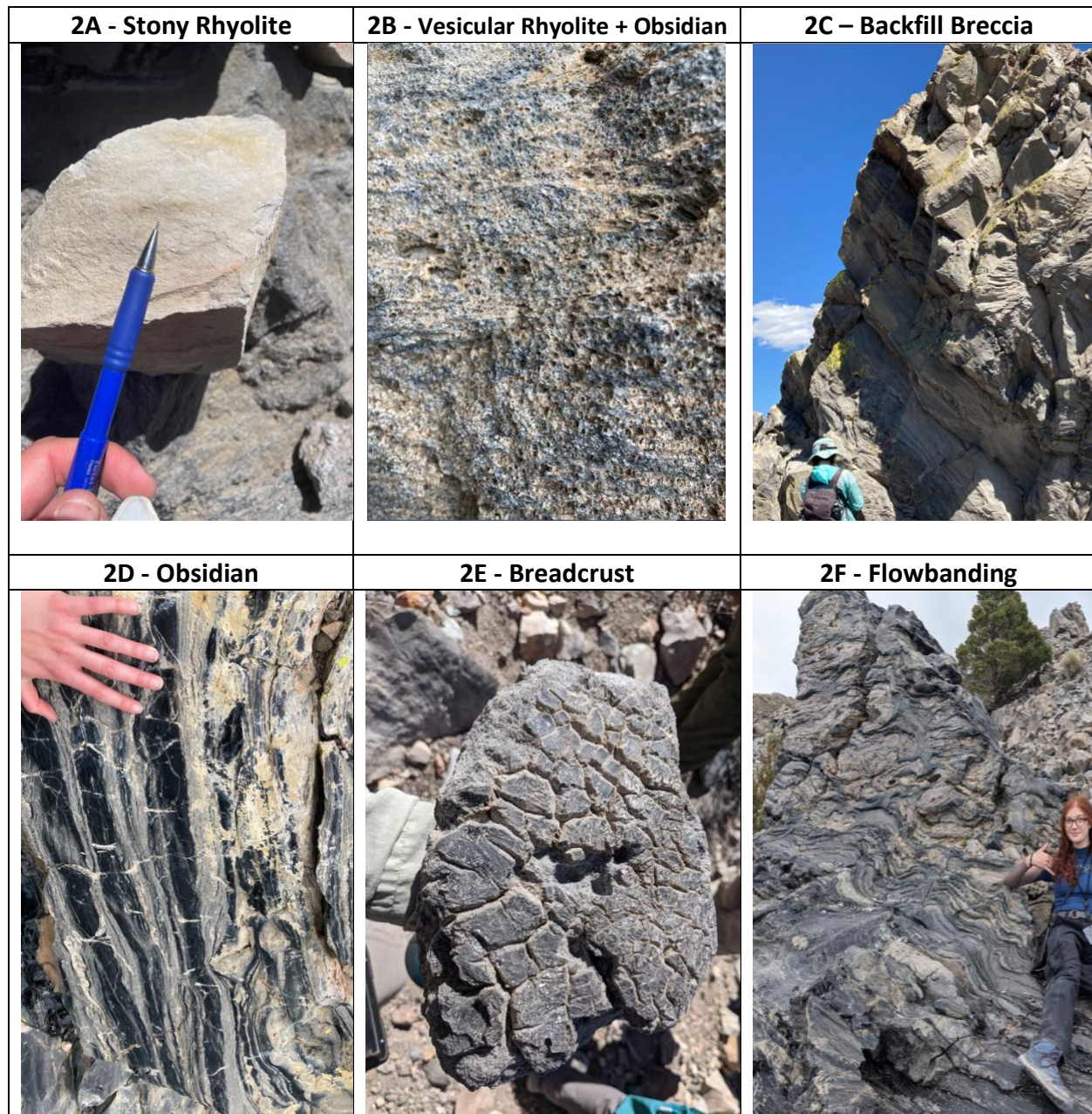
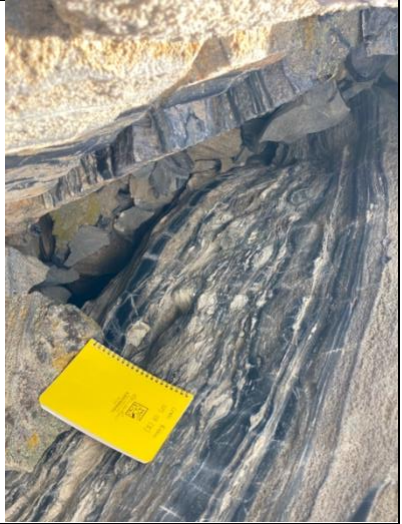


Figure 2 – Photo Appendix



2G - Pele's Tears	2H - Devitrification	2I - Boudinage
		
2J - Auto-brecciation	2K - Fumarolic Alteration	2L - Tuffisite
		

2M - Crease Structures	2N - Spatter/Agglutination	2O - Bombs trapped in flows
		
2P - Tephra Cone	2Q – En Echelon Shearing	
		

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